BreakHis Dataset Analysis for Breast Cancer Classification

1. Introduction

This project presents an end-to-end analysis of the BreakHis dataset for breast cancer classification using deep learning. The objective is to build a reproducible pipeline including data preprocessing, exploratory analysis, model training, and evaluation.

2. Dataset Description

The BreakHis dataset contains **7,909 histopathological images from 81 patients**, collected at four magnification levels: **40×, 100×, 200×, and 400×**.

Each image is labeled as:

- **Benign**
- **Malignant**

The dataset is imbalanced:

- Malignant: 5429 images
- Benign: 2480 images

```
Total images found: 7909
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/content/drive/MyDrive/BreakHis_Project/data/BreaKHis_v1/BreaKHis_v1/histology_slides/breast/malignant/SOB/ductal_carcinoma/SOB_M_DC_14-9461/100X/SOB_M_DC-14-9461-100-018.png
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```

Metadata rows: 7909

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2	/content/drive/MyDrive/BreakHis_Project/data/B...	SOB_M_DC-14-9461-100-019.png	SOB	malignant	1	DC	ductal_carcinoma	14-9461	100	019
3	/content/drive/MyDrive/BreakHis_Project/data/B...	SOB_M_DC-14-9461-100-018.png	SOB	malignant	1	DC	ductal_carcinoma	14-9461	100	018
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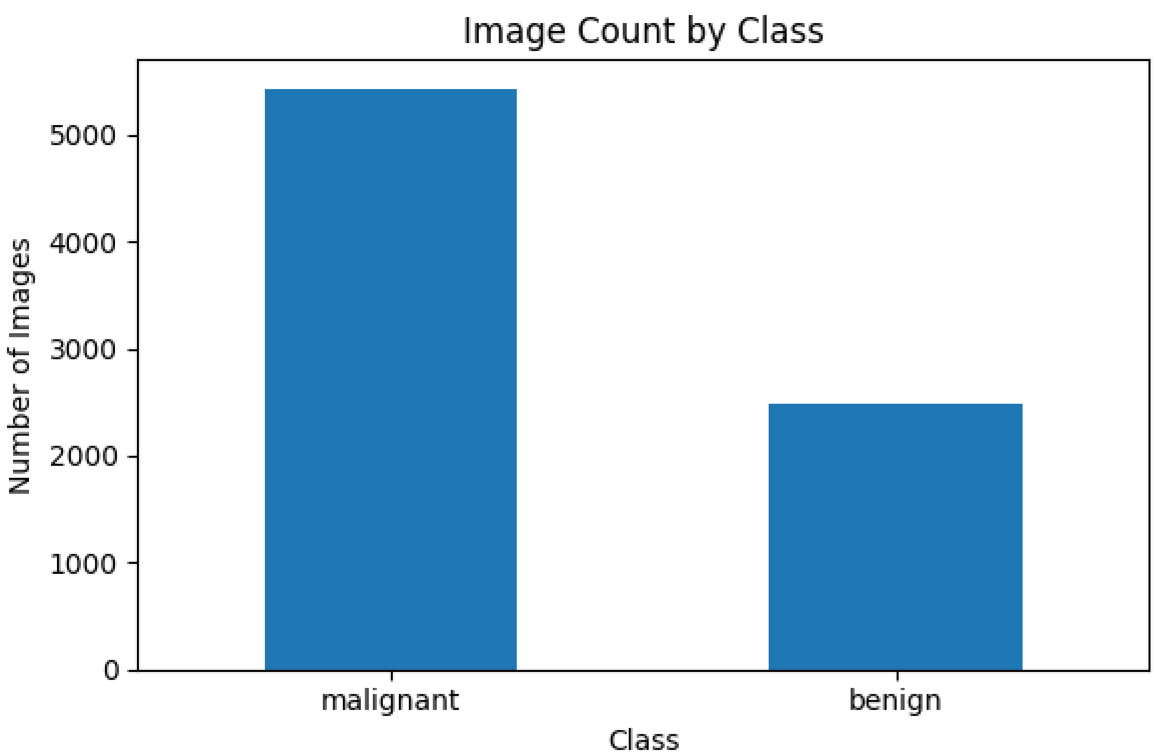
3. Methodology

Data Processing

- Extracted metadata from filenames
- Created structured dataset using pandas
- Encoded labels as binary values

Preprocessing

- Resized images to **224×224**
- Applied normalization using ImageNet statistics
- Applied augmentation (flip + rotation)



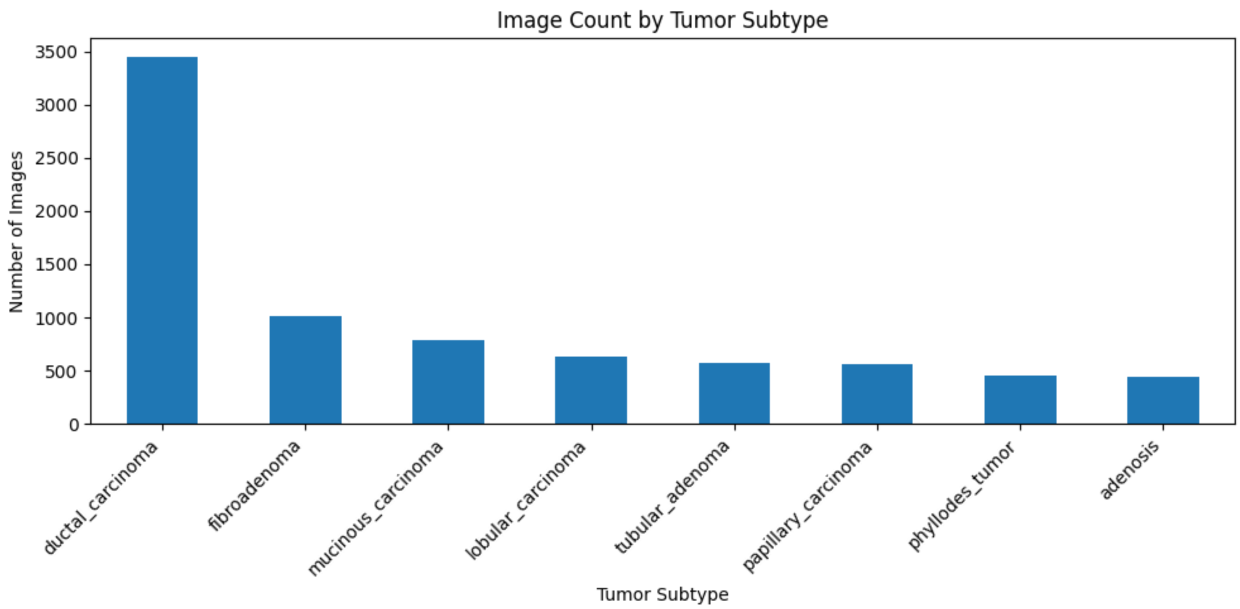


Image Count by Tumor Subtype
Number of Images
3500
3000
2500
2000
1500
1000
500
0
ductal_carcinoma
fibroadenoma
mucinous_carcinoma
lobular_carcinoma
tubular_adenoma
papillary_carcinoma
phyllodes_tumor
adenosis
Tumor Subtype

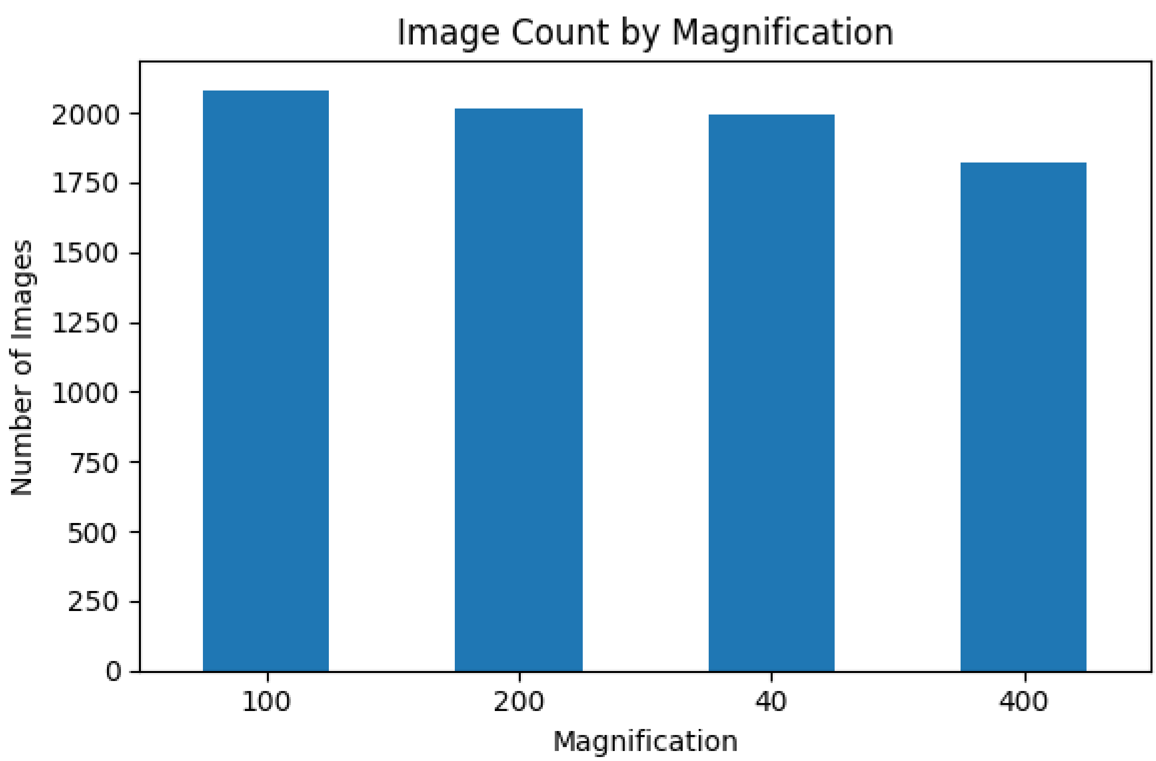
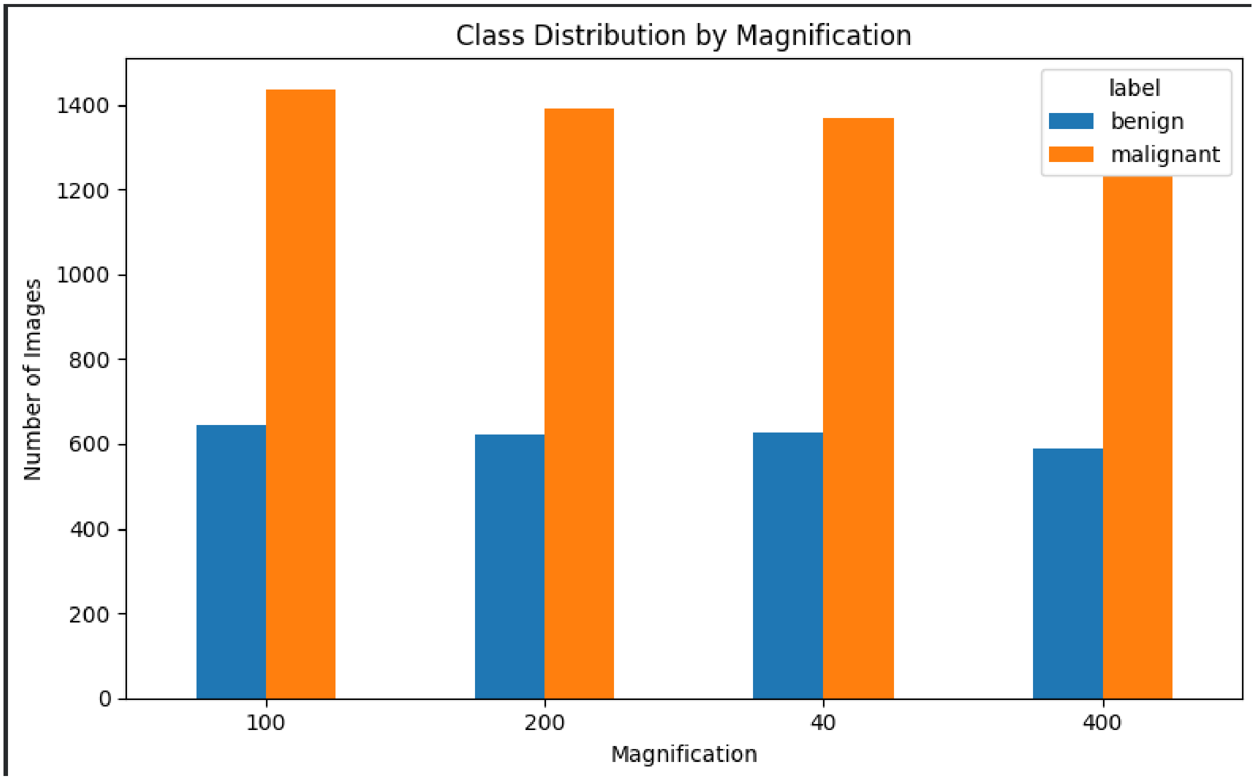
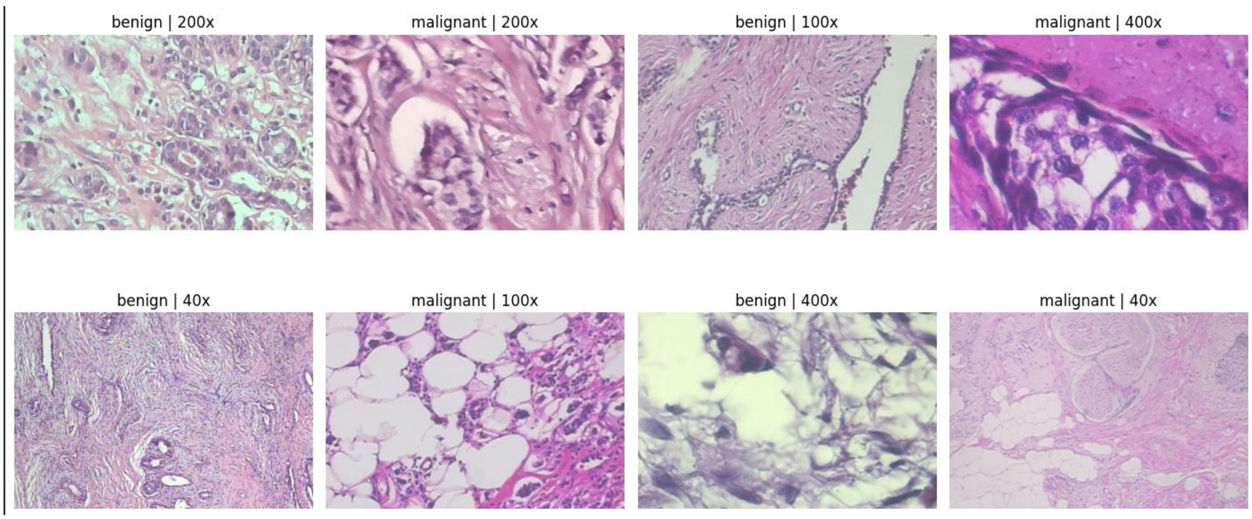


Image Count by Magnification
Number of Images
2000
1750
1500
1250
1000
750
500
250
0
100
200
40
400
Magnification





Data Splitting

- Used **patient-aware split** to avoid data leakage
- Train: 70%
- Validation: 15%
- Test: 15%

4. Exploratory Data Analysis

Key observations:

- Dataset is imbalanced toward malignant cases
- Subtypes are unevenly distributed (ductal carcinoma dominant)
- Magnification levels are relatively balanced
- Visual differences exist but are subtle

5. Model

A **MobileNetV2 transfer learning model** was used.

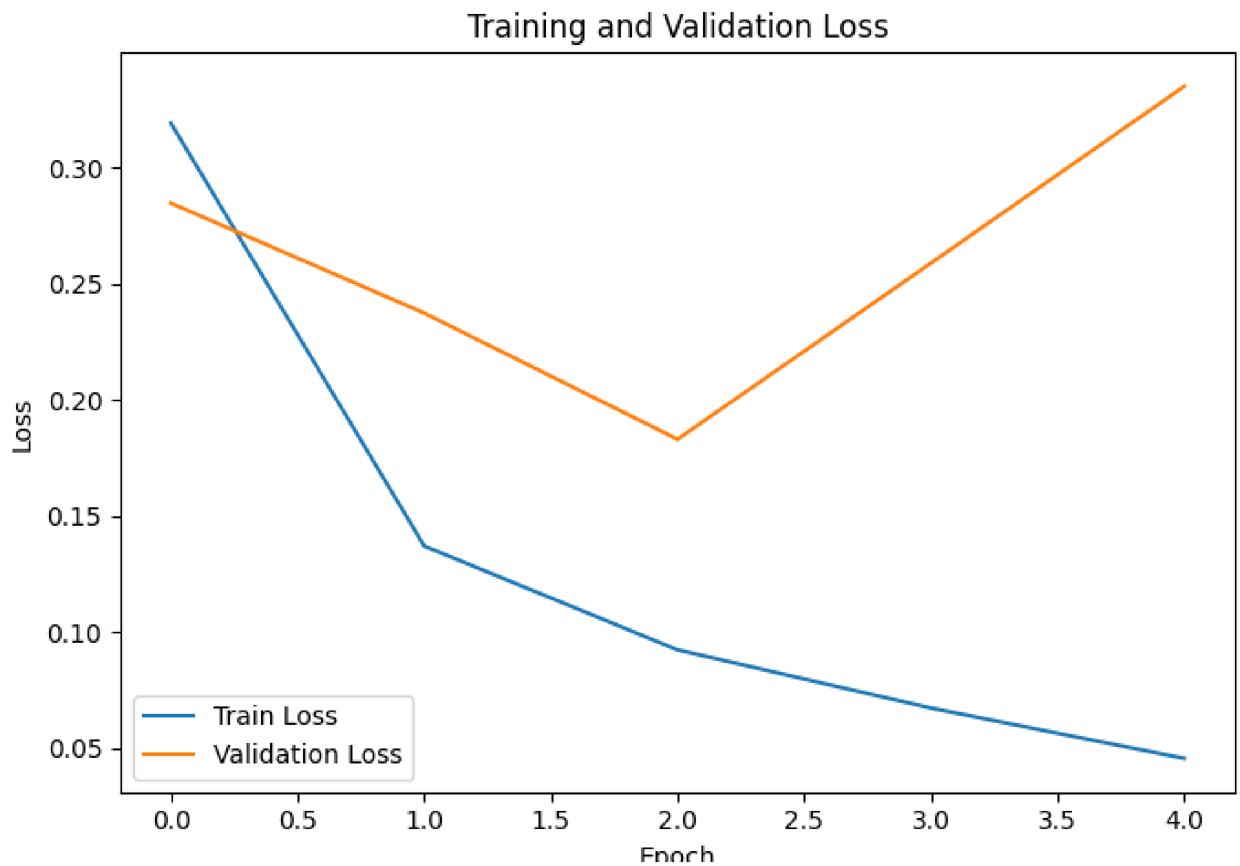
Training setup:

- Optimizer: Adam
- Learning rate: 0.0001
- Epochs: 5
- Loss: CrossEntropy

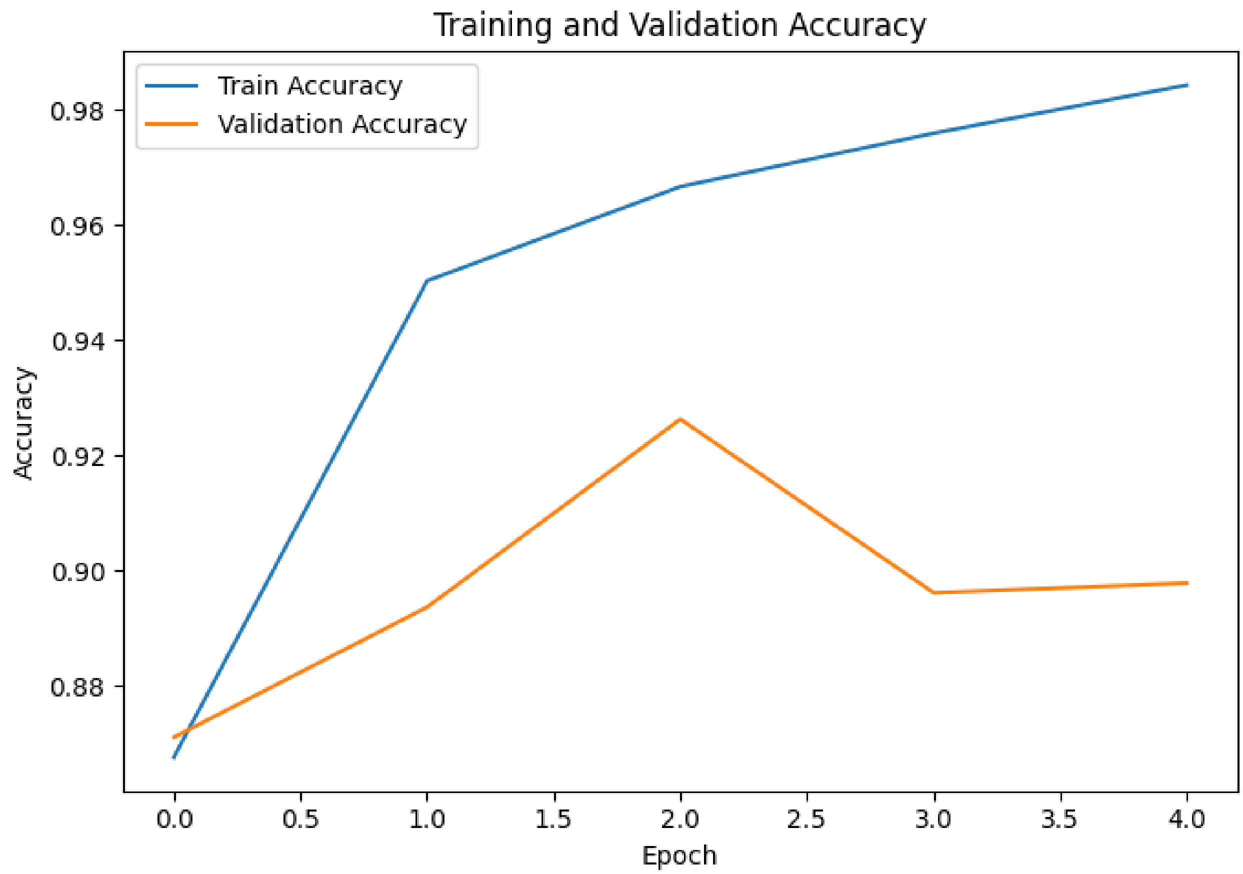
6. Results

Overall Performance

- Accuracy: **89.28%**
- Precision: **96.66%**
- Recall: **90.42%**
- F1-score: **93.43%**
- ROC-AUC: **0.947**



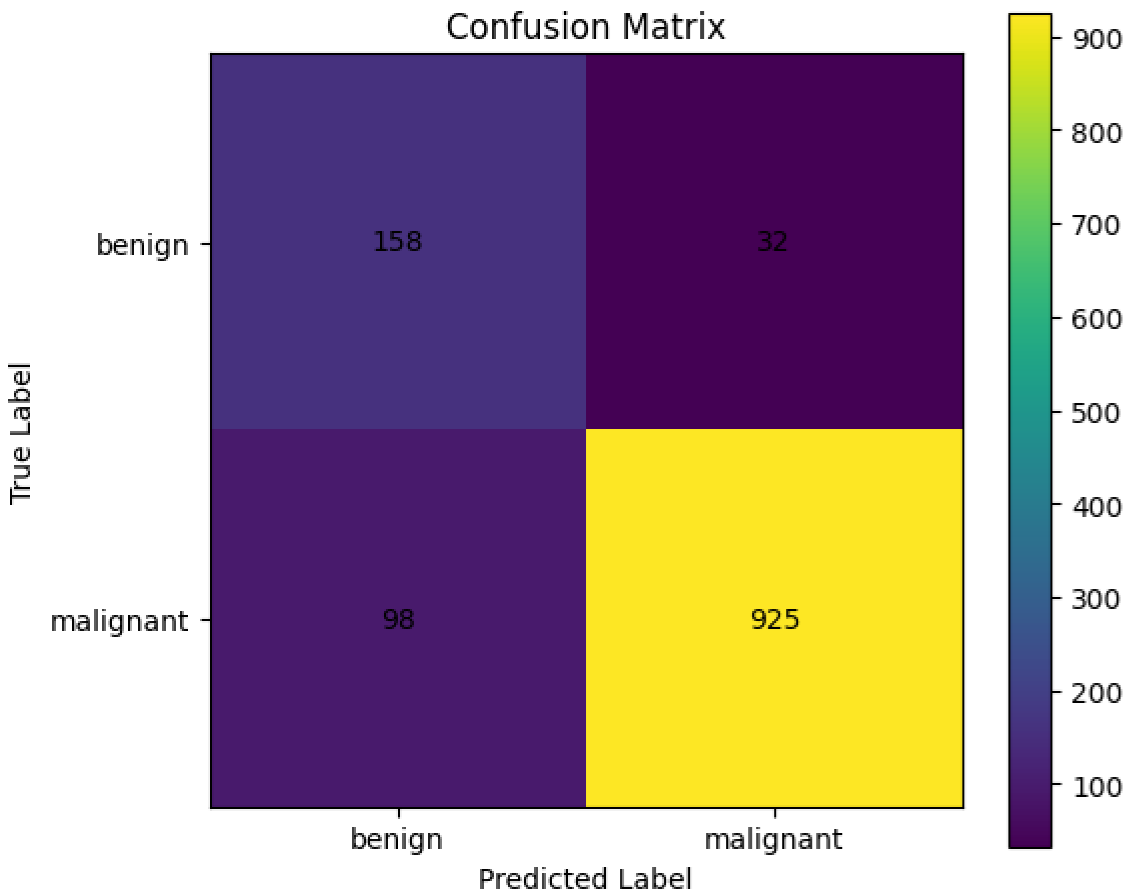
Training and Validation Loss
Loss
Epoch
Train Loss
Validation Loss
0.05
0.10
0.15
0.20
0.25
0.30
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0



Confusion Matrix Analysis

- False negatives (malignant → benign): 98
- False positives (benign → malignant): 32

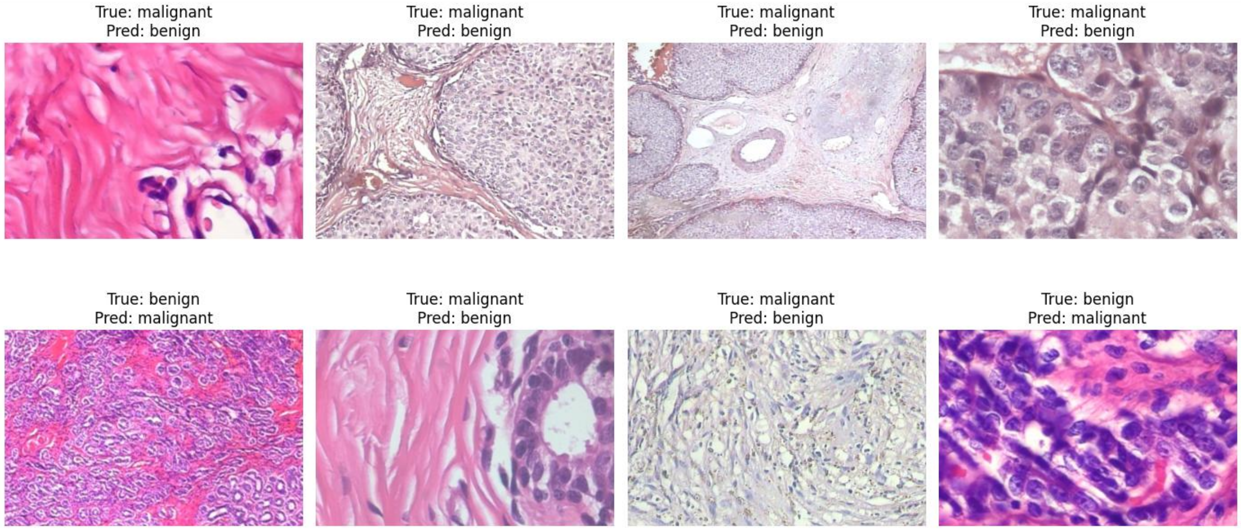
The model performs well overall but occasionally misclassifies malignant cases as benign, which is critical in medical applications.

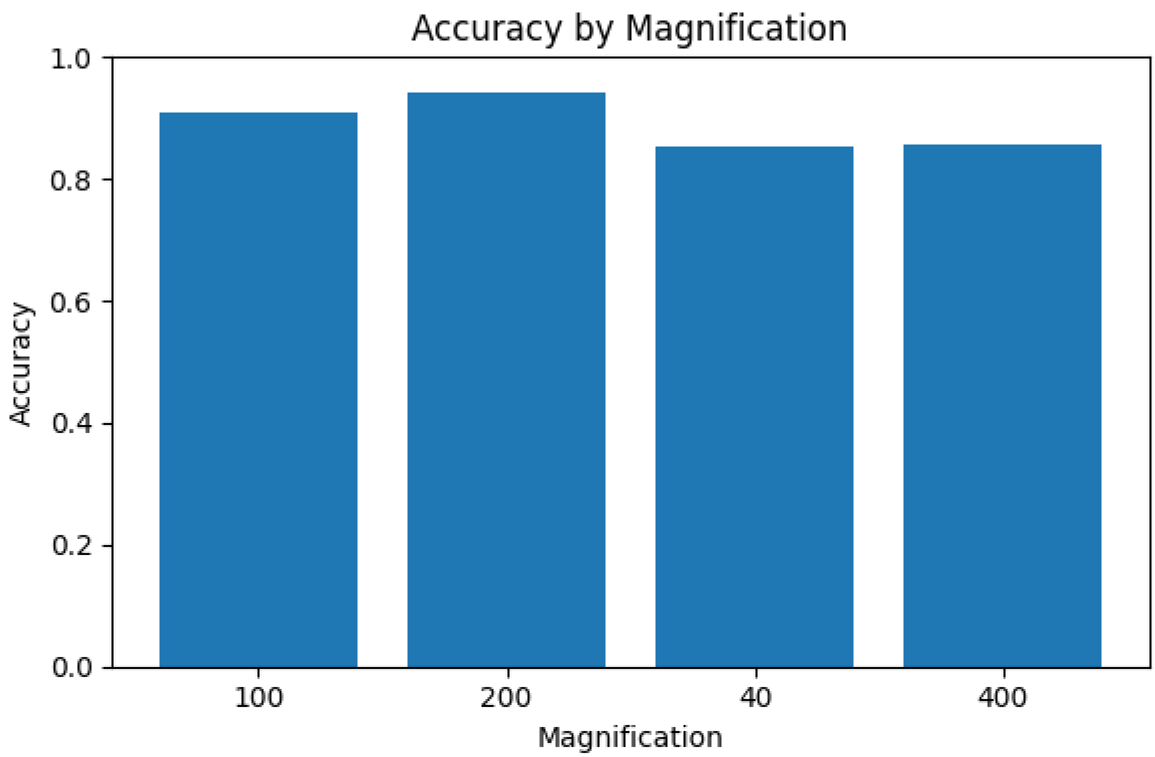


7. Performance by Magnification

- Best performance at **200× (94%)**
- Lower performance at **40× and 400× (~85–86%)**

This suggests that mid-level magnification provides optimal diagnostic information.





8. Error Analysis

- 130 misclassified images identified
- Errors often occur where visual patterns are ambiguous
- Some benign samples resemble malignant tissue and vice versa

9. Limitations

1. Dataset imbalance biases model toward malignant class
2. Limited number of patients (81)
3. Slight overfitting observed
4. No interpretability methods (e.g., Grad-CAM)
5. Not clinically validated

10. Conclusion

The model demonstrates strong performance for binary classification of histopathological images. However, due to dataset limitations and medical risks, results should not be considered clinically reliable.

11. Ethical Considerations

Medical AI systems must:

- Avoid overclaiming accuracy
- Be rigorously validated
- Prioritize minimizing false negatives

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===== FINAL SUMMARY =====
Total images: 7909
Unique patients/slides: 81
Train images: 5501
Validation images: 1195
Test images: 1213
Test accuracy: 0.8928276999175597
Precision: 0.9665621734587252
Recall: 0.9042033235581622
F1-score: 0.9343434343434344
ROC-AUC: 0.9470494417862839
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